

Data Visualisations 2026: Project

(50% of the total grade (equivalently 15 points))

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1 Key Dates

Handed out: March 17, 2026.

Submission of the project: **May 10, 2026 (within 23:59)**

2 Overview

Your group will design a visual project around a real-world question. The project must combine data analysis and exploration, interactive visualization, design reasoning, and critical reflection. The final result should be more than a collection of charts: it should help answer a clear set of questions.

This year, the project must include a **small original JavaScript component** and a **short paper-style report**. AI tools are allowed, but the project is designed so that you must **make and defend** your own analytical and design decisions.

3 Possible Topics

Choose a topic of social, scientific, economic, environmental, cultural, or policy relevance. The topic must be centered on a **question**, not just on a dataset. Example directions include:

- inequality, wages, education, and mobility;
- climate change, energy, pollution, and sustainability;
- health, demographics, migration, or wellbeing;
- tourism, transportation, housing, and urban development;
- social networks, institutions, or communication flows;
- gender, representation, and access to opportunities;
- AI in the job market; changes in skills demand, occupations, salaries, automation risk, or hiring trends;
- AI and ecology; energy use of AI systems, data centers, emissions, water consumption, e-waste, or environmental trade-offs.

In Section 8 you can check some possible datasets. *You may propose another topic if it is challenging / interesting from a visualisation point of view and supported by public data.*

Project Requirements

Each group must complete the following tasks:

1. Define **one central question**, **one target audience**, and **3–5 tasks** the system should support.
2. Use at least **two datasets** from public sources and discuss limitations and bias.

3. Submit at least one **Python notebook/ file** showing cleaning, integration, exploratory analysis, and the steps that help towards the design.
4. Build a visual analytics product with at least **six visual components** using Python, Microsoft Power BI, or Tableau (not necessarily in the same dashboard). If a group is with more than 3 people then at least 8 visual components.
 - At least **four visualizations must be interactive and linked**. The visualizations must allow for data filtering within themselves, preferably both by groups and by levels of detail/granularity
 - Include at least **one explanatory view** for a non-expert audience.
 - The visualizations must be meaningful and complex, maximum one line or bar chart, which should be styled with the same theme as the other charts, providing visual coherence. If you need to make a line chart consider possible variations like sparklines or inserting a marker on heightened moment.
 - **One "creative" visualisation**: it should present information in an innovative manner that deviates from standard visualization techniques. The objective is to design a visualization that not only effectively communicates your data but also engages and resonates with the audience by representing an idea or trend in a memorable way.
 - You will create two static visualizations of your dataset, each representing a different perspective: **white hat and black hat**. The white hat perspective focuses on ethical, transparent, and constructive insights, aiming to use data for positive impact. The black hat perspective explores how the same data could be misused, manipulated, or misleadingly presented, whether intentionally or through biased representation. You may use any visualization technique or tool, and the two visualizations do not need to follow the same approach.
5. Develop a **WordPress webpage** to present the project, embedding or linking the visualizations. The webpage should include a very brief introduction to the project, a description of the dataset and task analysis, and interactive visualizations. They should also include a conclusion that summarizes the main findings from the project.
6. Include at least **one small original JavaScript-based interactive feature** in the final webpage.

4 Paper-Style Report

In addition to the webpage, each group must submit a **maximum 5-page report** written as a short paper. The page limit excludes references and the AI appendix. The report should be concise and analytical, not a diary of everything you did.

Recommended structure:

1. **Introduction and question**: what matters and why.
2. **Data**: sources, integration, cleaning, and limitations.
3. **Audience, tasks, and design rationale**: who the system is for and why the chosen visual forms are appropriate.
4. **System description**: main views, interactions, and the JavaScript component.
5. **Rejected Alternatives** a small section where you explain at least two options you considered but did not use.
6. **Findings and limitations**: key insights, white-hat/black-hat reflection, and future improvements.

4.1 Tools

Use at least **three** of the following ecosystems: Python, Tableau, Power BI, and JavaScript.

5 AI Use

AI tools are encouraged to be used in this project, but only as **assistants**, not as substitutes for reasoning.

Allowed uses: brainstorming topics, finding documentation, debugging, learning syntax, improving wording, generating boilerplate code, or suggesting chart ideas.

Not allowed: inventing data or references, writing claims you cannot verify, generating large parts of the report without review, producing code you cannot explain, or using AI output as your final analysis without checking it.

Accountability rule: your group is responsible for every sentence, figure, code block, visual decision, and conclusion in the submission.

AI appendix (max 2 pages): In case you use AI, state which AI tools you used, for what purpose, 2–4 representative prompts (or prompt summaries), what you accepted or rejected, and one example where the AI output was weak, wrong, generic, or misleading and how you corrected it.

6 Final Deliverables

Submit one project package containing:

- Python notebooks and source code;
- the webpage and JavaScript files;
- Tableau / Power BI exports or links, if used;
- a **5-page paper-style report** in PDF;
- a README explaining how to run or access the project;
- an **AI use appendix** (max 2 page);

If the project is hosted online, include the public link in the README.

7 Presentation

Each group will give a **10-minute presentation** followed by questions. You should clearly explain the motivation, data, design logic, main findings, and the JavaScript component. You should include in your presentation a demo of your webpage. All group members should be able to explain the work, **including the role AI played in the process**.

8 Evaluation

Criterion	Weight
Problem framing, audience, and analytical tasks	10%
Data collection, preprocessing, integration, and limitations	10%
Quality of visualization design and analytical depth	20%
Interactivity and overall coherence of the system	15%
JavaScript component	10%
White-hat / black-hat reflection and ethics	10%
5-page paper-style report	10%
Presentation and oral defense	15%

Suggested Datasets

Some possible datasets that cover a variety of socially and scientifically relevant topics. *Feel free to use any other suitable dataset, including more creative or unconventional ones. The only requirement is that the chosen data must allow you to satisfy all the project requirements. For example, it should be possible to design both the white hat and the black hat visualisations.*

- Greenhouse Gas Emissions (1990–2018): The Organization for Economic Co-operation and Development (OECD) provides comprehensive emissions data categorized by pollutant type (e.g., carbon monoxide, methane) and sector (e.g., energy, agriculture). The dataset offers multiple filtering options for different perspectives.
- Gender Equality Indicators (1960–2017): The World Bank tracks key metrics such as fertility rate, literacy levels, employment rates, business ownership, and wage disparities to measure gender equality worldwide. The linked dataset curates a smaller subset of the overall set of gender indicators which you are welcome to use as well.
- Civilian Complaints Against NYPD Officers: This dataset, compiled by ProPublica, includes over 12,000 civilian complaints filed against the New York Police Department (NYPD). It contains demographic data of complainants and officers, categorized allegations, and case outcomes.
- Gentrification and Demographic Changes: Collected by BuzzFeed News, this dataset explores shifts in neighborhood demographics as wealthier populations move in, potentially displacing existing residents. The dataset includes extensive documentation on data processing and analysis.
- Airbnb Listings and Neighborhood Impact: This dataset, provided by Inside Airbnb, offers detailed data on Airbnb listings, including host details, pricing, availability, and neighborhood characteristics. It is useful for analyzing the impact of short-term rentals on housing markets and urban development.